NOAH QIAO

n3qiao@uwaterloo.ca | (647) 671-7908 | LinkedIn | Portfolio

EDUCATION

University of Waterloo | Waterloo, ON Sep 2022 - Apr 2028
Candidate for Bachelor of Applied Science in Mechatronics Engineering

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB, Java, SQL, VHDL; data structures & algorithms
Embedded Systems: STM32, ESP32, Arduino, Teensy 4.1, Raspberry Pi, FreeRTOS, Zephyr, ARM Cortex-M
Robotics & Controls: ROS 2, PID control, forward/inverse kinematics, trajectory planning, sensor fusion/EKF, OpenCV, MediaPipe
Tools & Interfaces: Git/GitHub, Linux, CMake, GDB, JTAG/SWD, interrupts, DMA, PWM, I2C, SPI, UART, CAN
Hardware & Mechanical: Altium, KiCad, LTspice, PCB design, soldering, ESD/surge protection, sensor integration, SolidWorks, AutoCAD, 3D printing, sheet metal design

RELEVANT EXPERIENCE

Borrum Energy Solutions Sep 2025 - Dec 2025
R&D Electrical - Mechatronics Engineering
- Developed a wind-turbine temperature/humidity monitoring unit using Teensy 4.1, RTOS task scheduling, sensor integration, and SD-card data logging.
- Designed a KiCad PCB with ESD and surge-transient protection for harsh outdoor deployment; collaborated with UofT and UBC on microwave-based ice-detection testing.
- Automated electronic component inventory using Python and vendor APIs, reducing manual data entry by ~85%.

Excelitas Jan 2025 - Apr 2025
Mechanical Engineering Co-op
- Led assembly and validation of a next-generation power testing station, supporting design approval and production readiness.
- Redesigned test fixtures to increase testing efficiency by 300% and reduce production time.
- Created sheet-metal models and manufacturing drawings in SolidWorks and revised PCB connector layouts/schematics in Altium.

Sheartak Tools Ltd Sep 2023 - Dec 2023
Mechanical Engineering Associate
- Designed 10+ SolidWorks cutter-head profiles and detailed manufacturing drawings for customer machine upgrades and product expansion.
- Supported machine-upgrade design iterations, refining cutter-head concepts to improve woodcutting quality and customer usability.

Magna International Jan 2023 - Apr 2023
Test Technician
- Conducted validation testing on 10,000+ customer-supplied products; recorded measurements and reported outliers under ISO 17025 procedures.
- Assisted senior engineers with fixture design and test setup development to support product validation and repeatable measurements.

SELECTED ENGINEERING PROJECTS

12-DOF Quadruped Robotic Dog Jul 2026
- Designing and building a 12-DOF quadruped with 12 high-torque servos, Arduino, and PCA9685 PWM control; developing embedded C++ firmware for joint calibration and coordinated crawl-gait sequencing.
- Designed a 12 V power-distribution and voltage-regulation architecture for high-current servo actuation and stable embedded electronics.
- Integrating an ESP32-S3 camera, IMU, ultrasonic sensing, and wireless communication for computer vision, obstacle detection, orientation sensing, and autonomous behavior.

Autonomous Search & Rescue Robot Apr 2026
- Developed an autonomous robot for line following, object detection, pickup, transport, and return-to-start; implemented and tuned PID feedback control using real-time IR sensor input for stable line tracking.
- Integrated Raspberry Pi vision, Arduino control, servo actuation, serial communication, and a custom 3D-printed vacuum pickup system, achieving ~90% pickup success.

Gesture-Controlled Robotic Hand Aug 2024
- Designed and 3D printed a SolidWorks robotic hand; implemented Arduino motor control and real-time Python/OpenCV/MediaPipe gesture tracking for finger actuation.